

Coding Solution and Functional Features

FitBuddy AI Fitness Plan Generator

A feature-level description of the implemented FitBuddy workflow.

Main user flow

- 1 Open the home page and enter name, user ID, age, weight in kilograms, goal and intensity.
- 2 Submit the form; the server validates input and requests a workout plan plus nutrition or recovery tip.
- 3 Review all seven days, the current AI mode and the safety note.
- 4 Send feedback using the same user ID to revise the latest saved plan.
- 5 Retrieve the plan history through the page or JSON API.

Feature inventory

Feature	Behavior documented in the guide
Plan creation	Creates a personalized seven-day plan and stores it with the profile.
Demo generation	Provides deterministic sample output when demo generation is allowed.
Gemini generation	Uses Google's GenAI SDK when configured with a private API key and available model.
Feedback revision	Stores feedback and the current revision separately from the original plan.
Coach dashboard	Lists saved records and supports deletion behind the admin token.
JSON API	Exposes health, plan creation, retrieval, feedback and admin operations.

Plan history

The guide describes existing user IDs as profile lookups, not proof of identity. A new generation creates a separate plan record. The original plan, revised plan and latest feedback can be distinguished in storage.